

Explainable Foundation Models for Crop Type Classification & LLM-Powered Agricultural Reporting

A two-part framework combining EO foundation model benchmarking with Explainable AI, and proof-of-concept integration with Large Language Models for interpretable, scalable crop monitoring in data-scarce regions.

Part 1

FM + XAI Research

Part 2

LLM Integration

Goal

Practical & Usable

PART 1 — SCIENTIFIC CORE

Foundation Models for Crop Classification + XAI

Benchmarking whether pretrained EO foundation models can enable accurate crop type mapping with minimal ground truth, and using Explainable AI to reveal *what* these models learn and *why* they make their predictions.

FOUNDATION MODELS TO BENCHMARK

Prithvi-E0-2.0 NASA/IBM · 300M & 600M · HLS data · Temporal + location embeddings	SatMAE Temporal MAE · Multi-spectral · Sentinel-2 pretrained
ScaleMAE Scale-aware pretraining · GSD-adaptive · Multi-resolution	Baseline: Raw Features NDVI/spectral time series · No foundation model · Control benchmark

ADAPTATION TECHNIQUES — FROM HEAVY TO LIGHT

- Full Fine-tuning
- LoRA / Adapters
- Linear Probing
- Sklearn on Embeddings ★
- k-NN

Core experiment: extract embeddings from frozen FM → classify with SVM / Random Forest / Logistic Regression. Compare learning curves (5 → 200 samples per class) against raw-feature baselines. How few labels can we use while still getting acceptable accuracy?

EXPLAINABLE AI — INTERPRETING THE BLACK BOX

 SHAP / Permutation Feature importance on embeddings & raw bands	 Attention Maps Which patches & dates does the ViT attend to?	 Integrated Gradients Attribution back to spectral bands & timesteps
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XAI as the Connective Thread

In Part 1, XAI is a **research tool** — revealing which spectral bands, temporal windows, and spatial patterns drive crop discrimination. In Part 2, these same XAI outputs become the **content** that enables the LLM to produce grounded, trustworthy reports. XAI bridges spatial intelligence and natural language.

Key question: Do FM-based and raw-feature classifiers agree on which temporal/spectral features matter? Agreement validates; disagreement

ALTERNATIVE EO + LLM APPROACHES CONSIDERED

- A. End-to-End MLLM

Fuse EO encoder + LLM natively (EarthGPT pattern). Powerful but requires massive aligned datasets and generic visual encoders lack spectral depth.
- B. LLM-as-Agent

LLM orchestrates EO tools autonomously (GeoLLM-Squad). Flexible but assumes pre-built task-specific models exist.
- C. XAI-Enriched Pipeline ✓

FM embeddings → lightweight classifier → XAI → LLM reporting. Chosen: each component does what it's best at, buildable, interpretable.

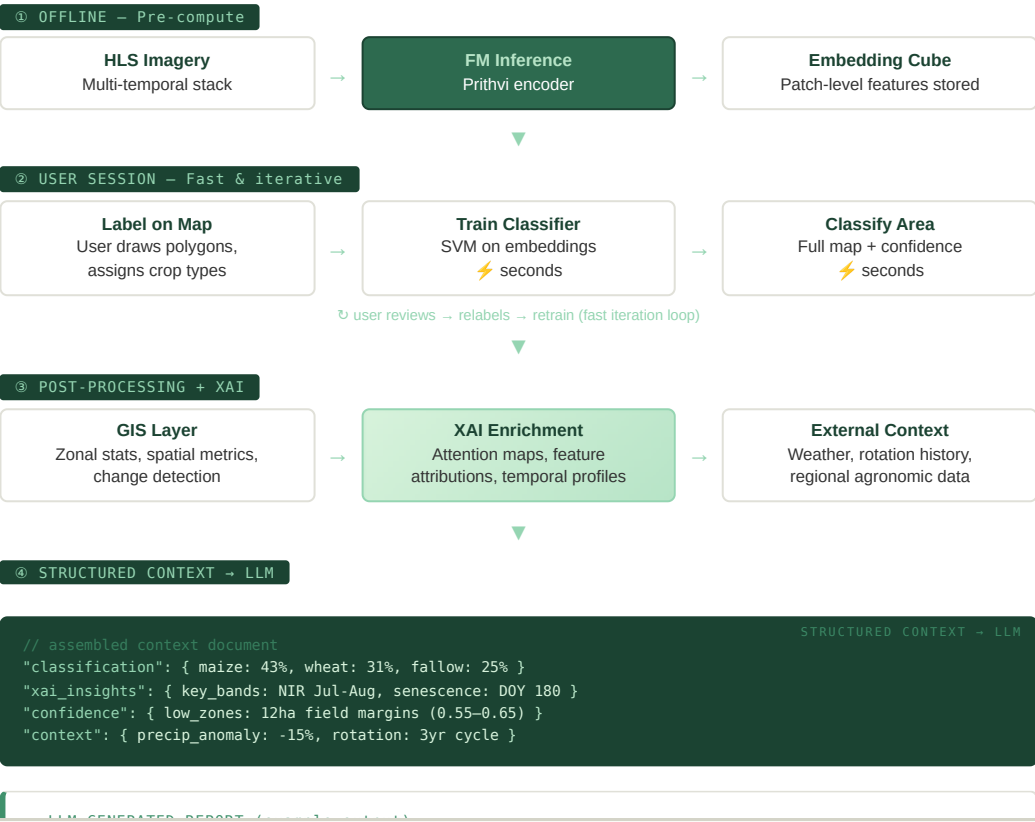
WHY THIS ARCHITECTURE

- No GPU at interaction time** — embeddings pre-computed; SVM trains in seconds on CPU.
- Fast user iteration** — relabel and retrain without re-running the FM.
- XAI bridges the gap** — transforms abstract FM embeddings into human-interpretable features that give the LLM grounded reasoning.
- Modular** — swap FM, classifier, or LLM independently.

PART 2 — APPLIED PROOF-OF-CONCEPT

LLM Integration for Agricultural Reports

Demonstrating how XAI-enriched crop classification outputs can feed a Large Language Model to produce interpretable, actionable agricultural monitoring reports — creating a usable end-to-end workflow.



CORE RESEARCH QUESTIONS

- How few labeled samples suffice for crop classification on FM embeddings vs raw features?
- Which spectral/temporal features do FMs learn to exploit — and do they align with agronomic knowledge?
- Can XAI outputs serve as effective context for LLM-generated agricultural intelligence?